Unit 2.2 Printables: Teacher Answer Key

Same order as the student printables. Where students write in a table, the table is filled in (answers in bold italic). Reference tables have nothing to fill in.

UNIT 2.2 · DAY 1 · GLUE IN · WRITE ON IT · KEY **Puppy litter data**

	Fur color	Ear shape	Pattern
Mom	Black	Floppy	Spotted
Dad	Brown	Pointed	Solid
Puppy A	Black (M)	Floppy (M)	Solid (D)
Puppy B	Brown	Pointed	Spotted
Puppy C	Black	Pointed	Spotted
Puppy D	Brown	Floppy	Solid

KEY Puppy A: M, M, D (done). Puppy B: fur D, ears D, pattern M. Puppy C: fur M, ears D, pattern M. Puppy D: fur D, ears M, pattern D. Every puppy matches one parent for each trait, but each has a different mix.

UNIT 2.2 · DAY 2 · GLUE IN · WRITE ON IT · KEY **Card table: 3 offspring**

Draw one card from each of the six piles. Write what you drew, then put the cards back. Offspring 1 is done for you.

Offspring	Fur color (P1 + P2)	Ear shape (P1 + P2)	Pattern (P1 + P2)
1 (done for you)	Black + White	Floppy + Pointed	Spotted + Solid
2			
3			

KEY Answers vary: each offspring combines one card per pile (one fur color from each parent, and so on).

UNIT 2.2 · DAY 4 · GLUE IN · WRITE ON IT · KEY **Genotype to phenotype: pea plants**

Genotype	Phenotype
TT	Tall (done for you)
Tt	***Tall***
tt	***Short***

KEY Tt: tall. tt: short.

UNIT 2.2 · DAY 4 · GLUE IN · COPY OF THE BOARD · KEY

Punnett squares: steps and worked example

1. Write one parent's two alleles across the top. **2.** Write the other parent's two alleles down the side. **3.** Fill each box with the allele from its column and the allele from its row.

Worked example: Tt × Tt

	T	**t**
T	TT	Tt
t	Tt	tt

Result: TT, Tt, Tt, tt. That is **3 tall to 1 short**.

Each box in a Punnett square shows one possible ____ of alleles.

KEY Answer: combination. Students keep this sheet for Days 4 and 6.

UNIT 2.2 · DAY 5 · GLUE IN · REFERENCE AND WRITE ON IT · KEY

Monster Genetics: the coin rule and Table 2

Trait	**Allele 1 (heads)**	**Allele 2 (tails)**
Eyes	Two small eyes (E)	One large eye (e)
Eye color	Red (R)	White (r)
Skin color	Green (G)	Blue (g)

Trait	**Genotype**	**Phenotype**
Eyes	ee	One large eye (done for you)
Eye color	***rr***	White
Skin color	Gg	***Green***
Tail shape	***cc***	Straight
Tail color	Pp	***Purple***

KEY Table 2 answers: Eye color rr; Skin color green; Tail shape cc; Tail color purple.

UNIT 2.2 · DAY 6 · GLUE IN · WRITE ON IT · KEY **Punnett square: skin color and eyes**

	G	**g**
G	GG	Gg
g	Gg	gg

Eyes: *ee* × *ee*

	e	**e**
e	ee	ee
e	ee	ee

KEY Eyes: ee × ee gives ee in all four boxes (100% one large eye).

UNIT 2.2 · DAY 6 · GLUE IN · REFERENCE · KEY **Prediction vs. class count**

Baby's skin color	**Punnett prediction**	**Class count (sample)**	**Class percent**
Green	75%	22 of 30	73%
Blue	25%	8 of 30	27%

KEY Reference table (sample class). Use your class's tally if you like.

UNIT 2.2 · DAY 7 · GLUE IN · REFERENCE · KEY **UV light and mutations**

Hours of UV light	**Mutations per 1,000 skin cells**
0 (none)	2
1	9
2	21
4	40

KEY Circle the 4-hour row (40 mutations per 1,000 cells).

UNIT 2.2 · DAY 7 · GLUE IN · WRITE ON IT · KEY

Who passes it on? Five cases

Case	Where the DNA changed	Inherited?
A	UV hits a skin cell	No (done)
B	Copy error in a sperm-making cell	***Yes***
C	Copy error in a muscle cell	***No***
D	X-rays hit an egg-making cell	***Yes***
E	Smoke chemical hits a lung cell	***No***

KEY A No; B Yes; C No; D Yes; E No.

UNIT 2.2 · DAY 8 · GLUE IN · REFERENCE · KEY

Case data: three cases

Case 1: puppy litter

	Alleles for fur color (B = black, b = brown)
Mom	B and b
Dad	B and b
Puppies	4 black (BB or Bb), 2 brown (bb)

Case 2: DNA of a puppy and its parents

	Allele 1	Allele 2
Mom	CAT	CAT
Dad	CAT	CGT
Puppy	CAT	CCT

Case 3: X-rays and mice

Group of mice	X-ray dose	New mutations per 1,000 sperm cells
A	none	3
B	low	12
C	high	25

KEY Reference tables. Case 2: the puppy's CCT is in neither parent (a mutation). Case 3: none = 3 (copying errors), high = 25, so the X-rays added about 22.

UNIT 2.2 · DAY 9 · GLUE IN · REFERENCE · KEY

The new coat pattern: two kennels

Kennel	Puppies checked	Puppies with a new coat pattern (in neither parent)
A (chemical in the water)	1,000	14
B (clean water)	1,000	3

KEY Reference table. 14 − 3 = 11 extra new patterns per 1,000 puppies in Kennel A.

UNIT 2.2 · DAY 10 · GLUE IN · WRITE ON IT · KEY

My target tracker

Write your CFA score out of 5. Below a B (fewer than 4 of 5)? Write Yes under Review. At the end of class, rate each target 1 to 4 (4 = I could teach it).

Target	CFA score (of 5)	Review?	Rating (1 to 4)
LS3-2.1 three sources			
LS3-2.2 sex cells and fertilization			
LS3-2.3 copying errors and UV			
LS3-2.4 read data, make a claim			

KEY Student scores and ratings; nothing to key.

UNIT 2.2 · DAY 10 · GLUE IN · REFERENCE · KEY

Hydrangea soil data

Soil pH	Flower color (same DNA)
5.0	Blue
6.0	Purple
7.0	Pink

KEY Reference table (optional enrichment).